CHAPTER B

Fluency bank

Two questions per session, in 6 to 8 minutes including correction. The identifiers allow items to be revisited; they are not a list to be completed in full in every lesson. The availability dates are based on the planned teaching; postpone them if the topic has not yet been studied.

R001 · Fractions

Calculate $\frac{2}{3}+\frac{1}{4}$. **Solution.** $\frac{11}{12}$: common denominator 12.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R002 · Order of operations

Calculate $5-2(3-7)$. **Solution.** 13: brackets, product, difference.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R003 · Powers

Simplify $\dfrac{10^{5}\times 10^{-3}}{10^{4}}$. **Solution.** $10^{-2}=0{,}01$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R004 · Roots

Simplify $\sqrt{50}$. **Solution.** $5\sqrt{2}$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R005 · Proportion

18 represents 30% of a total. What total? **Solution.** $\dfrac{18}{0{,}3}=60$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R006 · Direct rate

A price of 80 falls by 15%. Final value? **Solution.** $80\times 0{,}85=68$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R007 · Initial value

After an increase of 20%, the value is 72. Initial value? **Solution.** $\dfrac{72}{1{,}2} = 60.$

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R008 · Calculated rate

A quantity goes from 40 to 50. Rate? **Solution.** $\dfrac{50-40}{40} = 25\,\%.$

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R009 · Successive rates

20% increase then 20% decrease: overall rate? **Solution.** $1{,}2 \times 0{,}8 = 0{,}96$: a 4% decrease.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R010 · Reverse rate

What rate offsets a 20% decrease? **Solution.** $\dfrac{1}{0{,}8} = 1{,}25$: a 25% increase.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R011 · Zero product

Solve (2 $x-1)(x+3)=0$. **Solution.** $x=\dfrac{1}{2}$ or $x=-3$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R012 · Expanding

Expand (2 $x-3)^2$. **Solution.** $4x^2-12x+9$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R013 · Factorising

Factorise 3 x^2-12x. **Solution.** $3x(x-4)$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R014 · Sign of an affine function

Solve $-3x + 6 \geq 0$. **Solution.** $x \leq 2$: reverse the inequality when dividing by -3.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R015 · Gradient

Slope of the line through (1; 4) and (3; 0)? **Solution.** -2.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R016 · Line

Equation of the line with slope -2 through (1; 4)? **Solution.** $y = -2x + 6$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R017 · Weighted average

The values 2 and 5 have frequencies 3 and 2. Mean? **Solution.** $\dfrac{6 + 10}{5} = 3{,}2$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R018 · Median

Median of 1, 2, 2, 5, 8, 10? **Solution.** $\dfrac{2 + 5}{2} = 3{,}5$, using the convention of averaging the two middle values.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R019 · Quartiles

For the data set 1, 2, 3, 4, 5, 6, 7, 8, give Q_1 and Q_3 with the ceiling indices $\left\lceil \dfrac{n}{4} \right\rceil$ and $\left\lceil \dfrac{3n}{4} \right\rceil$. **Solution.** $Q_1 = 2$, $Q_3 = 6$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R020 · Statistical standard deviation

For 1 and 3 equally weighted: mean, variance and standard deviation? **Solution.** Mean 2, variance $\frac{[(1-2)^2+(3-2)^2]}{2}$ = 1, standard deviation 1.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R021 · Units

Convert 1 min 40 s into seconds; 2400 frames over this duration give how many per second? **Solution.** 100 s; 24 frames/s.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R022 · Simple probability

With a fair die, what is the probability of a multiple of 3? **Solution.** $\frac{2}{6}=\frac{1}{3}$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R023 · Complement

$P(A)=\frac{3}{8}$. Give $P(\overline{A})$. **Solution.** $\frac{5}{8}$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R024 · Interval

Write $x \geq -2$ and $x < 3$ in interval notation. **Solution.** $[-2;3[$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R025 · Logic

Negate "all the pupils passed". **Solution.** "At least one pupil did not pass."

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R026 · Histogram

Two classes with widths 2 and 4 have the same frequency, 20. In a histogram with areas proportional to the frequencies, what are the relative heights? **Solution.** Heights 10 and 5: the area, not the height alone, encodes the frequency.

Prerequisites: Prior knowledge

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.

R027 · Pie chart

A sector measures 72 °. Proportion? **Solution.** $\frac{72}{360} = 20\,\%$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R028 · Scale

An axis starts at 90. Bars end at 100 and 110. Is the second value double the first? **Solution.** No: $\frac{110}{100} = 1{,}1$. The heights drawn above 90 make this comparison misleading.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R029 · Percentage of a percentage

40% of pupils do A; among them, 25% do B. What proportion do both A and B? **Solution.** $0{,}4 \times 0{,}25 = 10\,\%$.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R030 · Proportion and frequency

One category makes up 0.125 of 80 people. How many people? **Solution.** 10 people.

Prerequisites: Prior knowledge

Available from 2026-09-03; suggested reviews: 2026-09-10, 2026-09-24, 2026-11-03.

R031 · Defining and representing a sequence

Calculate the first four terms of $w_0 = 2,\ w_{n+1} = w_n + 2n + 1$; suggest an explicit formula. **Solution.** $2, 3, 6, 11$. The formula $w_n = n^2 + 2$ works: it gives the first term and $(n+1)^2 + 2 - (n^2+2) = 2n + 1$.

Prerequisites: Algebraic manipulation, integer powers, percentages, affine functions, Python loops.

Available from 2026-09-09; suggested reviews: 2026-09-17, 2026-10-01, 2026-11-03.

R032 · Arithmetic sequences

We know $u_3 = 17$ and $u_8 = 37$ for an arithmetic sequence. Determine r and u_0. **Solution.** $37 - 17 = 5r$, so $r = 4$ then $u_0 = 17 - 3 \times 4 = 5$.

Prerequisites: Algebraic manipulation, integer powers, percentages, affine functions, Python loops.

Available from 2026-09-11; suggested reviews: 2026-09-22, 2026-10-06, 2026-11-03.

R033 · Geometric sequences

Study the variation of $u_n = -2 \times 3^n$, then of $v_n = 2(-1)^n$. **Solution.** $u_{n+1} - u_n = -4 \times 3^n < 0$: decreasing. v_n alternates 2 and -2: neither increasing nor decreasing.

Prerequisites: Algebraic manipulation, integer powers, percentages, affine functions, Python loops.

Available from 2026-09-16; suggested reviews: 2026-09-24, 2026-10-08, 2026-11-05.

R034 · Sums, subscripts and proofs

Calculate $\sum_{k=2}^{5} 2^k$ without a summation symbol in the answer. **Solution.** $4 + 8 + 16 + 32 = 60$, or $\dfrac{4(1-2^4)}{1-2} = 60$.

Prerequisites: Algebraic manipulation, integer powers, percentages, affine functions, Python loops.

Available from 2026-09-18; suggested reviews: 2026-09-29, 2026-10-13, 2026-11-10.

R035 · Conjecturing a limit and programming a threshold

For $u_n = 48(\frac{1}{-})^n$, determine the first indices where $u_n \leq 6$ and where $u_n < 6$. **Solution.** $u_3 = 6$: index 3 for ≤ 6, index 4 for < 6.

Prerequisites: Algebraic manipulation, integer powers, percentages, affine functions, Python loops.

Available from 2026-09-18; suggested reviews: 2026-09-29, 2026-10-13, 2026-11-10.

R036 · Three forms, three readings

Give three forms of $g(x) = x^2 + 2x - 3$. Read its roots and minimum. **Solution.** $g(x) = (x+1)^2 - 4 = (x-1)(x+3)$. Roots 1 and -3, minimum -4 at -1.

Prerequisites: Special products, zero-product property, sign of a product, square roots.

Available from 2026-09-23; suggested reviews: 2026-10-01, 2026-10-15, 2026-11-12.

R037 · Discriminant and solving

Solve 3 $x^2 - 6x + 3 = 0$ and $x^2 + 2x + 5 = 0$. **Solution.** The first is 3($x - 1)^2 = 0$, so 1. For the second, ?= $-16 < 0$: no real solution.

Prerequisites: Special products, zero-product property, sign of a product, square roots.

Available from 2026-09-25; suggested reviews: 2026-10-06, 2026-11-03, 2026-11-17.

R038 · Factoring without automating the discriminant

Determine the quadratic polynomial f such that $f(1) = f(4) = 0$ and $f(0) = -8$. **Solution.** $f(x) = a(x-1)(x-4)$; $4a = -8$, so $f(x) = -2(x-1)(x-4)$.

Prerequisites: Special products, zero-product property, sign of a product, square roots.

Available from 2026-09-30; suggested reviews: 2026-10-08, 2026-11-03, 2026-11-19.

R039 · Sign, inequalities and optimisation

Solve $x^2 - 4x - 5 < 0$, then maximise $x(8 - x)$ on [0; 3]. **Solution.** $(x - 5)(x + 1) < 0$ gives $]-1; 5[$. The vertex of $x(8-x)$ is at 4, outside the domain; on [0; 3], the function increases and its maximum is 15 at 3.

Prerequisites: Special products, zero-product property, sign of a product, square roots.

Available from 2026-10-02; suggested reviews: 2026-10-13, 2026-11-03, 2026-11-24.

R040 · Projection and angle

Two vectors have norms 2 and 5 and form an angle of 90 °, then 180 °. Calculate the scalar products. **Solution.** 0 then -10. The cosines are respectively 0 and -1.

Prerequisites: Vectors, Chasles' relation, Pythagoras, cosine in a right triangle, orthonormal coordinates.

Available from 2026-10-07; suggested reviews: 2026-10-15, 2026-11-03, 2026-11-26.

R041 · Coordinates, norm and orthogonality

For which real t are the vectors (t; 2) and (3; -6) orthogonal? **Solution.** $3t - 12 = 0$, therefore $t = 4$.

Prerequisites: Vectors, Chasles' relation, Pythagoras, cosine in a right triangle, orthonormal coordinates.

Available from 2026-10-09; suggested reviews: 2026-11-03, 2026-11-03, 2026-12-01.

R042 · Vector calculation and identities

We are given $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$. **Solution.** The square equals $4 - 2 + 9 = 11$; the norm is $\sqrt{11}$.

Prerequisites: Vectors, Chasles' relation, Pythagoras, cosine in a right triangle, orthonormal coordinates.

Available from 2026-10-14; suggested reviews: 2026-11-03, 2026-11-05, 2026-12-03.

R043 · Al-Kashi and choice of method

A triangle has $AB = AC = 4$ and $BC = 4\sqrt{2}$. Determine angle A. **Solution.** $32 = 16 + 16 - 32\cos A$, so cos $A = 0$ and $A = 90°$.

Prerequisites: Vectors, Chasles' relation, Pythagoras, cosine in a right triangle, orthonormal coordinates.

Available from 2026-10-16; suggested reviews: 2026-11-03, 2026-11-10, 2026-12-08.

R044 · The circle with diameter AB

For $A(0; 0)$, $B(6; 0)$, find the points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$. **Solution.** $MI^2 - 9 = 7$: circle with centre (3; 0) and radius 4.

Prerequisites: Vectors, Chasles' relation, Pythagoras, cosine in a right triangle, orthonormal coordinates.

Available from 2026-10-16; suggested reviews: 2026-11-03, 2026-11-10, 2026-12-08.

R045 · Rate of change and secants

Calculate the rate of $f(x) = x^2 + 3x$ between 2 and 2 + h. **Solution.** $f(2+h) - f(2) = 7h + h^2$; rate $7 + h$ for $h \neq 0$.

Prerequisites: Images, slope of a line, algebraic manipulation, square root and absolute value.

Available from 2026-11-04; suggested reviews: 2026-11-12, 2026-11-26, 2027-01-05.

R046 · Derivative at a point

For $f(x) = 3x^2 - 2x$, calculate $f\prime(1)$ from the rate. **Solution.** The rate is 4 + 3 h, so the derivative is 4.

Prerequisites: Images, slope of a line, algebraic manipulation, square root and absolute value.

Available from 2026-11-06; suggested reviews: 2026-11-17, 2026-12-01, 2027-01-05.

R047 · Tangent and affine approximation

We are given $f(3) = 5$ and $f\prime(3) = -2$. Write the tangent and approach $f(3{,}04)$. **Solution.** $y = 5 - 2(x-3) = -2x + 11$; approximation $5 - 2(0{,}04) = 4{,}92$.

Prerequisites: Images, slope of a line, algebraic manipulation, square root and absolute value.

Available from 2026-11-11; suggested reviews: 2026-11-19, 2026-12-03, 2027-01-05.

R048 · When differentiability fails

Justify that $g(x) = |x-2|$ is not differentiable at 2. Also study the differentiability of $x \mapsto \sqrt{x}$ at 0 using its rate of change. **Solution.** The rate of change at 2 is $\dfrac{|h|}{h}$, with opposite one-sided limits. For the square root at 0, the rate $\dfrac{1}{\sqrt{h}}$ for $h > 0$ exceeds every bound as h approaches 0: no real derivative value.

Prerequisites: Images, slope of a line, algebraic manipulation, square root and absolute value.

Available from 2026-11-13; suggested reviews: 2026-11-24, 2026-12-08, 2027-01-05.

R049 · Conditioning: changing the denominator

Out of 60 objects, 20 are red; of the red ones, 5 are damaged. Calculate $P(R \cap D)$ and $P_R(D)$. **Solution.** $\dfrac{5}{60} = \dfrac{1}{12}$ and $\dfrac{5}{20} = \dfrac{1}{4}$.

Prerequisites: Fractions, proportions, probability on a finite sample space, intersection and union.

Available from 2026-11-18; suggested reviews: 2026-11-26, 2026-12-10, 2027-01-07.

R050 · Trees, partitions and total probabilities

$P(A) = \dfrac{1}{3}$, $P_A(B) = \dfrac{3}{4}$, $P_{\overline{A}}(B) = \dfrac{1}{4}$. Calculate $P(B)$. **Solution.** $\dfrac{1}{4} + \dfrac{1}{6} = \dfrac{5}{12}$.

Prerequisites: Fractions, proportions, probability on a finite sample space, intersection and union.

Available from 2026-11-20; suggested reviews: 2026-12-01, 2026-12-15, 2027-01-12.

R051 · Independence and Incompatibility

We are given $P(A) = 0{,}4$, $P(B) = 0{,}5$, $P(A \cap B) = 0{,}18$. Are they independent? **Solution.** No: $0{,}4 \times 0{,}5 = 0{,}20 \neq 0{,}18$.

Prerequisites: Fractions, proportions, probability on a finite sample space, intersection and union.

Available from 2026-11-25; suggested reviews: 2026-12-03, 2026-12-17, 2027-01-19.

R052 · Repeated trials and Bernoulli

Four independent trials with probability $p = \frac{1}{3}$. Calculate the probability of four failures, then of at least one success. **Solution.** $(\frac{2}{3})^4 = \frac{16}{81}$ then $1 - \frac{16}{81} = \frac{65}{81}$.

Prerequisites: Fractions, proportions, probability on a finite sample space, intersection and union.

Available from 2026-11-27; suggested reviews: 2026-12-08, 2027-01-05, 2027-01-19.

R053 · Simulate and discuss the model

A simulation of 2000 points gives 675 points under $y = x^2$. What area does it estimate? **Solution.** $\frac{675}{2000} = 0{,}3375$; this is an estimate, not the exact value.

Prerequisites: Fractions, proportions, probability on a finite sample space, intersection and union.

Available from 2026-11-27; suggested reviews: 2026-12-08, 2027-01-05, 2027-01-19.

R054 · Derivative function and standard functions

Find d/dx x^5, x^{-2} and $\sqrt{x}$ stating the domains. **Solution.** $5x^4$ on $\mathbb{R}$; $-\frac{2}{x^3}$ for $x \neq 0$; $\frac{1}{2\sqrt{x}}$ for $x > 0$.

Prerequisites: Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

Available from 2026-12-02; suggested reviews: 2026-12-10, 2027-01-05, 2027-01-21.

R055 · Sum, product, reciprocal and quotient

Differentiate $(x^2+1)(3x-2)$ then $\frac{1}{x^2+1}$. **Solution.** $2x(3x-2)+3(x^2+1) = 9x^2-4x+3$; $-\frac{2x}{(x^2+1)^2}$, on $\mathbb{R}$.

Prerequisites: Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

Available from 2026-12-04; suggested reviews: 2026-12-15, 2027-01-05, 2027-01-26.

R056 · Sign of the derivative and extrema

Study $f(x) = x^2 - 4x + 1$ on [0; 3] and give its extrema over this interval. **Solution.** $f' = 2x - 4$, so decreasing up to 2, then increasing. Minimum -3 at 2; values at the endpoints 1 and -2, so maximum 1 at 0.

Prerequisites: Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

Available from 2026-12-09; suggested reviews: 2026-12-17, 2027-01-05, 2027-01-28.

R057 · Parity, inequalities and comparison

Study the parity of $x^4 + x^2$ and $x^3 - x$; prove $x^2 + 1 \geq 2x$. **Solution.** The first is even, the second odd; difference $x^2 + 1 - 2x = (x-1)^2 \geq 0$, equality at 1.

Prerequisites: Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

Available from 2026-12-11; suggested reviews: 2027-01-05, 2027-01-05, 2027-02-02.

R058 · Optimisation and Newton's method

A rectangle with perimeter 12 m has width x. Find the maximum area using differentiation. **Solution.** $A(x) = x(6-x)$ on [0; 6]; $A\prime = 6 - 2x$. Maximum 9 m ^{2} at 3.

Prerequisites: Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

Available from 2026-12-11; suggested reviews: 2027-01-05, 2027-01-05, 2027-02-02.

R059 · Normal vector and Cartesian equation

Equation of the line through (2; −1) with normal vector (3; 4). **Solution.** $3(x-2) + 4(y+1) = 0$, i.e. $3x + 4y - 2 = 0$.

Prerequisites: Vector coordinates, scalar product, linear systems, completing the square. Coordinate system always orthonormal.

Available from 2027-01-06; suggested reviews: 2027-01-19, 2027-01-28, 2027-03-02.

R060 · Orthogonal projection and distance

Project $P(4; 1)$ onto $d : x = 2$. **Solution.** $H(2; 1)$ and distance 2; the normal vector is horizontal.

Prerequisites: Vector coordinates, scalar product, linear systems, completing the square. Coordinate system always orthonormal.

Available from 2027-01-08; suggested reviews: 2027-01-19, 2027-02-02, 2027-03-02.

R061 · Equation of a circle

Identify $x^2 + y^2 + 2x - 8y + 17 = 0$. **Solution.** $(x+1)^2 + (y-4)^2 = 0$: the single point $(-1; 4)$, not a circle with strictly positive radius.

Prerequisites: Vector coordinates, scalar product, linear systems, completing the square. Coordinate system always orthonormal.

Available from 2027-01-13; suggested reviews: 2027-01-21, 2027-02-04, 2027-03-04.

R062 · Solve a configuration

Intersect $(x-1)^2 + (y+2)^2 = 9$ with $y = 1$. **Solution.** $(x-1)^2 = 0$, so only the point (1; 1); the line is tangent to the circle.

Prerequisites: Vector coordinates, scalar product, linear systems, completing the square. Coordinate system always orthonormal.

Available from 2027-01-15; suggested reviews: 2027-01-26, 2027-02-09, 2027-03-09.

R063 · Variable, event and distribution

Two fair coins are tossed. X counts the tails. Write its distribution and $P(X \leq 1)$. **Solution.** Values 0,1,2; probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. $P(X \leq 1) = \frac{3}{4}$.

Prerequisites: Probabilities, weighted mean, squares, trees and lists.

Available from 2027-01-20; suggested reviews: 2027-01-28, 2027-02-11, 2027-03-11.

R064 · Expectation and linearity

A gross gain is 10 with probability $\frac{1}{5}$ and 0 otherwise. What stake makes the game fair? **Solution.** $E(G) = \frac{10}{5}$ = 2. For a stake m, $E(G - m) = 2 - m$; fair stake 2.

Prerequisites: Probabilities, weighted mean, squares, trees and lists.

Available from 2027-01-22; suggested reviews: 2027-02-02, 2027-03-02, 2027-03-16.

R065 · Variance, standard deviation and König·Huygens

For X with values -1, 1 equally likely, calculate the expectation, variance and standard deviation. **Solution.** $E(X) = 0$, $E(X^2)$ = 1, variance 1 and standard deviation 1.

Prerequisites: Probabilities, weighted mean, squares, trees and lists.

Available from 2027-01-27; suggested reviews: 2027-02-04, 2027-03-02, 2027-03-18.

R066 · Transforming and comparing winnings

$E(X) = 3$, $V(X)$ = 4. Give the expectation and variance of $Y = -2X + 5$. **Solution.** $E(Y) = -1$, $V(Y) = 16$, $\sigma(Y) = 4$.

Prerequisites: Probabilities, weighted mean, squares, trees and lists.

Available from 2027-01-29; suggested reviews: 2027-02-09, 2027-03-02, 2027-03-23.

R067 · Samples and observed mean

For $\mu = 5$, σ = 3 and n=36, give the requested experimental interval. **Solution.** $\frac{2\sigma}{\sqrt{n}} = 1$; interval [4; 6].

Prerequisites: Probabilities, weighted mean, squares, trees and lists.

Available from 2027-01-29; recommended repeats: 2027-02-09, 2027-03-02, 2027-03-23.

R068 · Defining a new function

What is the equation of the tangent to e^x at 1? **Solution.** $y = e + e(x - 1) = ex$.

Prerequisites: Powers, geometric sequences, differentiation and sign.

Available from 2027-02-03; recommended repeats: 2027-02-11, 2027-03-02, 2027-03-25.

R069 · Algebraic properties and comparisons

Simplify $\dfrac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$. **Solution.** e^{2x+1} ; $x+2 > 2x$, therefore $x < 2$.

Prerequisites: Powers, geometric sequences, differentiation and sign.

Available from 2027-02-05; recommended repeats: 2027-03-02, 2027-03-02, 2027-03-30.

R070 · Differentiating and studying

Find g' if $g(t) = 3e^{-2t}$ and study how it varies. **Solution.** $g\prime(t) = -6e^{-2t} < 0$ on $\mathbb{R}$: strictly decreasing.

Prerequisites: Powers, geometric sequences, differentiation and sign.

Available from 2027-02-10; recommended repeats: 2027-03-02, 2027-03-04, 2027-04-06.

R071 · From discrete to continuous

For $Q(t) = 5e^{0,3t}$, express $Q(n)$ and $\dfrac{Q(t+2)}{Q(t)}$. **Solution.** $u_n = 5(e^{0,3})^n$; the ratio over two units is $e^{0,6}$.

Prerequisites: Powers, geometric sequences, differentiation and sign.

Available from 2027-02-12; recommended repeats: 2027-03-02, 2027-03-09, 2027-04-06.

R072 · Building numerically and proving an inequality

Show that $e^{2x} \geq 1 + 2x$ for all real x. **Solution.** Apply the proved inequality with argument 2 x ; equality only at x=0.

Prerequisites: Powers, geometric sequences, differentiation and sign.

Available from 2027-02-12; recommended repeats: 2027-03-02, 2027-03-09, 2027-04-06.

R073 · Measuring an angle in radians

Convert 45 °, 120° ; give the arc with radius 2 m and angle $\dfrac{3\pi}{4}$. **Solution.** $\dfrac{\pi}{4}, \dfrac{2\pi}{3}$; length $\dfrac{3\pi}{2}$ m.

Prerequisites: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-03; recommended repeats: 2027-03-11, 2027-03-25, 2027-04-27.

R074 · Wrapping and the image of a real number

Plot the images of $\dfrac{7\pi}{2}$ and $-\dfrac{5\pi}{3}$ reducing them to one turn. **Solution.** $\dfrac{7\pi}{2}$ has the same image as $\dfrac{3\pi}{2}$; $-\dfrac{5\pi}{3}$ that of $\dfrac{\pi}{3}$.

Prerequisites: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-05; recommended repeats: 2027-03-16, 2027-03-30, 2027-04-27.

R075 · Sine, cosine and standard values

If $\cos x = \frac{3}{5}$ and the image of x is in the fourth quadrant, give $\sin x$. **Solution.** $\sin^2 x = \frac{16}{25}$; the sine is negative, so $-\frac{4}{5}$.

Prerequisites: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-10; recommended repeats: 2027-03-18, 2027-04-06, 2027-04-29.

R076 · Associated angles and symmetries

Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $-\frac{\pi}{4}$. **Solution.** $(-\frac{1}{2}; -\frac{\sqrt{3}}{2})$ and $(\frac{\sqrt{2}}{2}; -\frac{\sqrt{2}}{2})$.

Prerequisites: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-12; recommended repeats: 2027-03-23, 2027-04-06, 2027-05-04.

R077 · Approximating pi and reinvesting

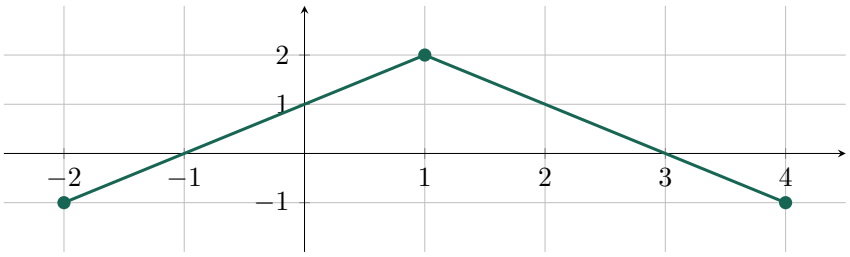
For n=6 and c=1, calculate the new chord after doubling. **Solution.** $h = \frac{\sqrt{3}}{2}$; new chord $\frac{1}{\sqrt{2+\sqrt{3}}} = \sqrt{2-\sqrt{3}}$.

Prerequisites: Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Available from 2027-03-12; recommended repeats: 2027-03-23, 2027-04-06, 2027-05-04.

R078 · Graphical reading

In Figure R078, read $f(1)$, solve $f(x) = 0$ and $f(x) > 0$.



Solution. $f(1) = 2$; zeros -1 and 3; positive sign on $]-1;3[$.

Prerequisites: Locating points

Available from 2026-09-03; recommended repeats: 2026-09-10, 2026-09-24, 2026-11-03.